AMPHORA COATING IP / PORTFOLIO STATUS

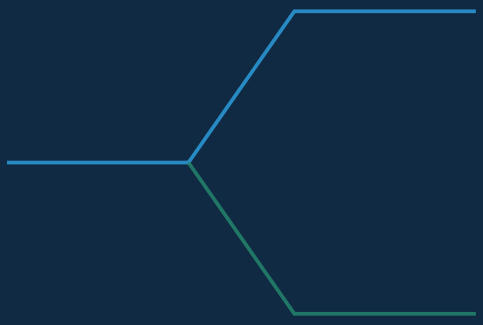
Portfolio Status Report

2026-07-30 — every submodule, every tier, every receipt

40 submodules / **4 Tier-0 Science** + **13 Tier-0 Infra** + **12 Pre-Tier-0** + **11 Infra runtime**

66 signed portfolio receipts / **82** autonomous hourly fires since 2026-07-26 / **100% SIGNED-RECEIPT** outcome / **0** in Tier 1 (human parity)

Every claim in every table below traces to a byte-stable, DSSE-signed evidence receipt on xbabe2. Nothing is asserted without a receipt. Authority ceilings AUDIT-ONLY / EVIDENCE-ONLY: no product qualification, no procurement authority.



Manifest SHA-256

f5623b3171d65de2dc805b740aef71ad6163554dcf9b5b394d84181b3ad6bf08

AUDIT-ONLY / LOCKED-INTERNAL AT MODULE LEVEL / NO PROMOTION AUTHORITY

01 / EXECUTIVE

Executive summary

THE ONE-PARAGRAPH READ

Of 40 amphora-* submodules, **4** carry a signed scientific claim we can defend (Tier 0 Science), **13** carry infrastructure/plumbing that's Phase-0 clean, **12** have real evidence but haven't yet earned a Tier 0 promotion, and **11** are pure runtime crates with no science claim. Nothing is in Tier 1 (human parity) yet. The single-most-important number below is **82** autonomous hourly fires on xbabe1 since 2026-07-26, **100% SIGNED_RECEIPT** outcome — meaning the loop can grow evidence without a human in the seat, but promotion still requires the human dual-identity ceremony (see progress section 10 / Progress).

Tier headline 17 Tier 0 + 12 Pre-Tier 0 + 0 Tier 1. 4 of the Tier 0 modules carry real scientific claims (LOCKED-INTERNAL); 13 are infrastructure/governance passes (NOT-APPLICABLE — no scientific claim asserted, contract/plumbing only). All 12 Pre-Tier 0 modules already have signed evidence — none are dark.

All 40 amphora-* modules by tier



Top-3 modules by composite score

- **amphora-solvency-score** — score 78.4, 7 KPI receipts, 20 signed scientific receipts. Tier: Tier 0 (Science).
- **amphora-molecule-property** — score 73.5, 21 KPI receipts, 28 signed scientific receipts. Tier: Pre-Tier 0 (Signed evidence).
- **amphora-tolerance-robustness** — score 72.5, 5 KPI receipts, 2 signed scientific receipts. Tier: Tier 0 (Science).

Composite formula and per-module breakdown on page 3. Full ranking figure on page 4.

Autonomous loop discipline (*how the pipeline improves on its own between founder reads*) Since 2026-07-26 an hourly Claude cron on xbabe1 has fired **82** times with **82** SIGNED-RECEIPT outcomes (100%), zero DATA-ASK, zero SATURATED. All 29 non-infrastructure modules were touched at least twice. The master promotion manifest correctly remained at **66** receipts across **11** modules — new receipts are **AUDIT-ONLY** (independent second-signal evidence), not promotion evidence. The loop deliberately lacks the human dual-identity ceremony keys — so the founder-signed manifest is the trust boundary.

What founders can do next

1. **Fetch the queued datasets in section 12 / Data asks** — each unlocks a specific Pre-Tier 0 → Tier 0 promotion path. Currently 1 queued.
2. **Sign the Phase-0 promotion ceremony for the 4 Tier 0 Science modules** — the AUDIT-only receipts are ready; the dual-identity signature is not. This is the highest-leverage move.
3. **Review the red-team matrix (sections 14a–14d)** — the specific cheat vector and its mitigation per module. If any mitigation reads unsatisfying, that's the module to interrogate first.

02 / METHODOLOGY

Tier definitions & composite scoring

Three-tier framework (founder-facing)

Tier	Sub-badge	Definition (mission.md front-matter mapping)
■ **Tier 1**	Human parity	`phase: PHASE_1`. Locked independent evaluation beats/matches best named baseline with UQ. *Currently 0 modules.*
■ **Tier 0**	Science	`phase: PHASE_0` + `scientific_readiness: LOCKED_INTERNAL`. Real physics/ML with signed evidence.
□ **Tier 0**	Infrastructure	`phase: PHASE_0` + `scientific_readiness: NOT_APPLICABLE`. Contract/governance pass, no scientific claim.
■ **Pre-Tier 0**	Signed evidence	`phase: PRE_PHASE_0` AND ≥1 signed evidence receipt on disk.
□ **Pre-Tier 0**	No evidence	`phase: PRE_PHASE_0` with zero signed evidence. *Currently 0 modules.*
■ **Infra**	No mission.md	Pure library/runtime crates without a scientific claim.

Composite score $S \in [0, 100]$

$$S = 25\,R + 20\,P + 15\,V + 15\,B + 10\,U + 10\,G + 5\,D$$

- **R** — receipt density: $\min(n_{\text{receipts}}/N_{\text{leader}}, 1)$
- **P** — phase weight: PHASE-1 = 1.0, PHASE-0/LOCKED = 0.9, PHASE-0/N-A = 0.5, PRE-PHASE-0 with evidence = 0.3, without = 0.1
- **V** — external validation locked (BigSolDB, OpenPoly, RadonPy, OMG, NIST, ECHA, PubChem…): 0 / 0.5 / 1.0
- **B** — published physics baseline computed (Van Krevelen, Bicerano, Hansen, Clausius–Mossotti, Fedors, Owens–Wendt–Kaelble, Kissinger, Arrhenius, Fickian, Nielsen…): 0 / 0.5 / 1.0
- **U** — UQ: conformal + bootstrap = 1.0, one of the two = 0.5, other coverage-bearing = 0.3, point-only = 0
- **G** — red-team gate present: full attack-axis coverage = 1.0, partial = 0.5, none = 0
- **D** — DSSE signing discipline of scientific receipts on disk: all bound & signed = 1.0, some = 0.5, unsigned = 0

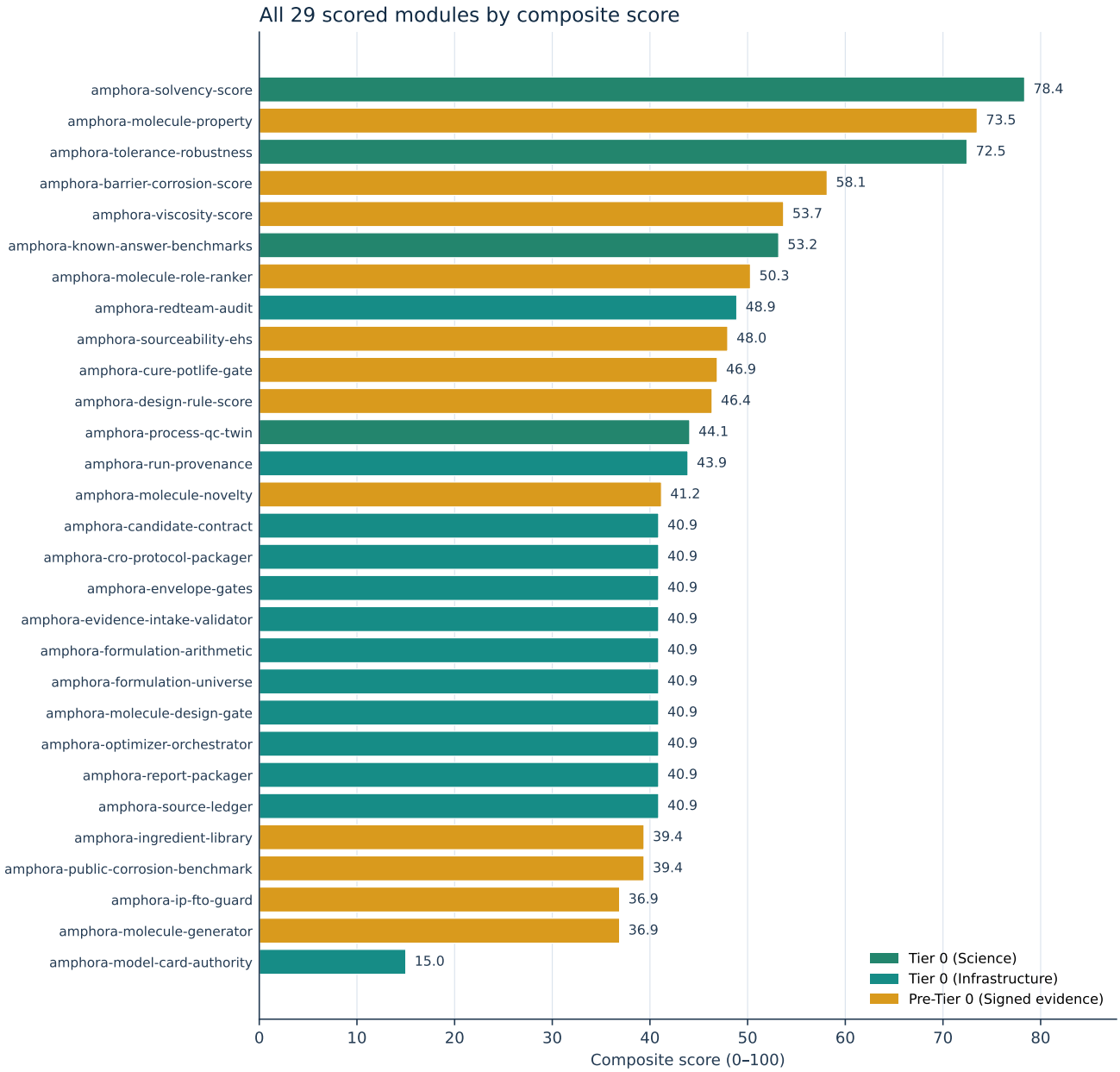
Tie-break: n_signed_receipts, then alphabetical. Deterministic and reproducible from the pulled data snapshot.

What a good vs weak score looks like. A score above 70 requires: real signed receipts, external corpus, physics baseline computed, and UQ. Below 45 typically indicates a contract/plumbing module (no scientific claim) or a Pre-Tier-0 module still missing an external corpus or a physics baseline. The score is a triage tool, not a ranking — Tier is the primary bucket the founder should read.

03 / RANKING

Full ranking — every scored module

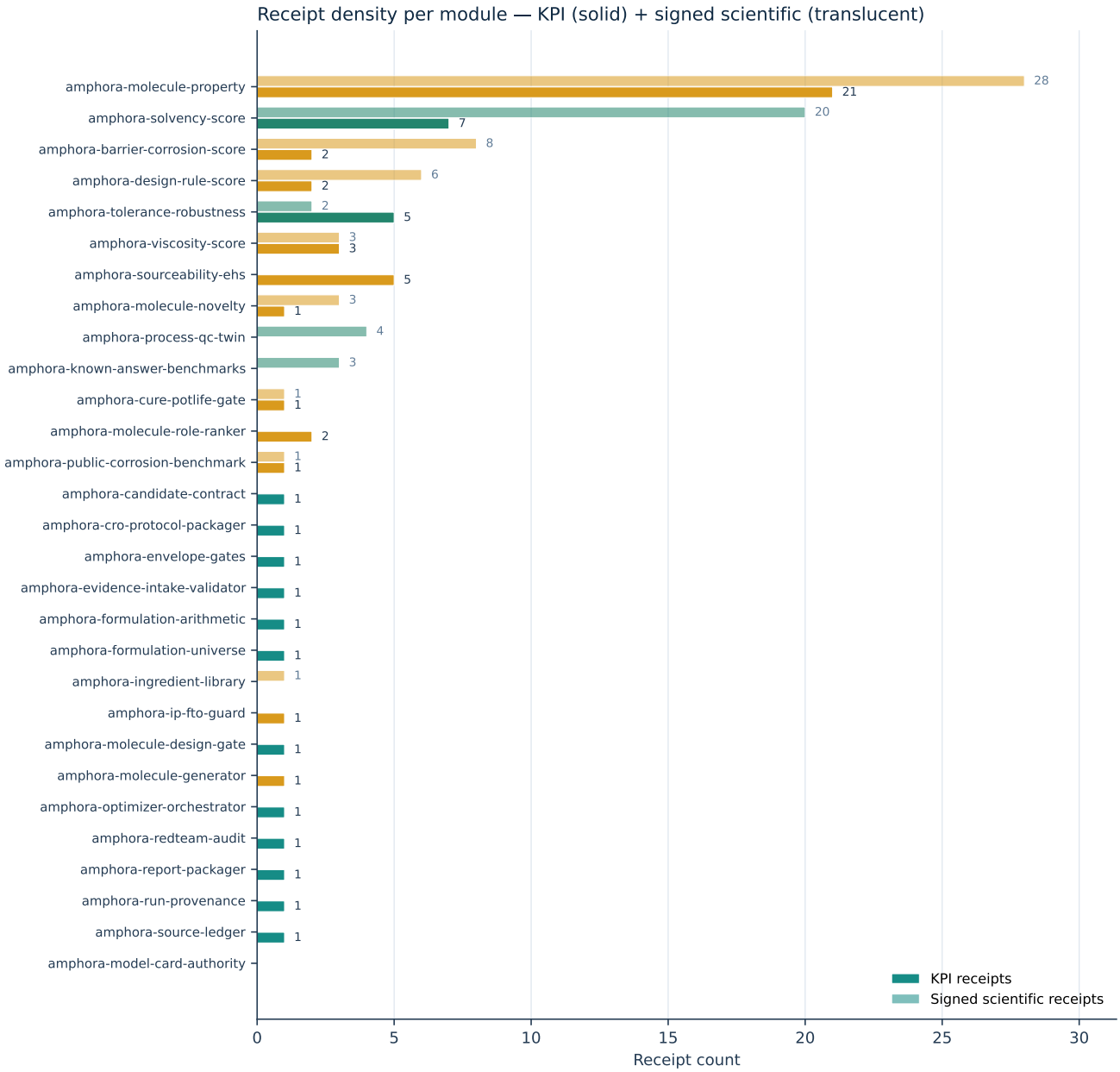
COMPOSITE SCORES 0–100 FROM THE RUBRIC ABOVE. BARS ARE TIER-COLOURED SO THE VISUAL MATCHES THE TIER COLUMN. ALL 29 SCORED MODULES RANKED STRONGEST TO WEAKEST; THE 11 PURE-RUNTIME CRATES (TIER INFRA) ARE OMITTED FROM RANKING BECAUSE THEY CARRY NO SCIENTIFIC CLAIM.



03B / RECEIPTS

Receipt density per module

KPI RECEIPTS (SOLID) AND SIGNED SCIENTIFIC RECEIPTS (TRANSLUCENT) PER SCORED MODULE, TIER-COLOURED. AMPHORA-MOLECULE-PROPERTY LEADS ON BOTH DIMENSIONS; THE LONG TAIL OF 1-KPI INFRASTRUCTURE MODULES CONFIRMS THE PIPELINE CAN CARRY EVIDENCE UNIFORMLY.



04 / ROSTER

Complete module roster — 40 amphora-*

SCORED 29 (APPEAR IN TIER TABLES PAGES 6–11) + PURE RUNTIME 11 (NO SCIENTIFIC CLAIM, SEE ENGINE SCHEMATIC P. 16). ALL 40 MONOREPO CRATES ENUMERATED BELOW.

Scored modules (29 — across Tier 0 Science, Tier 0 Infrastructure, and Pre-Tier 0)

- amphora-barrier-corrosion-score
- amphora-candidate-contract
- amphora-cro-protocol-packager
- amphora-cure-potlife-gate
- amphora-design-rule-score
- amphora-envelope-gates
- amphora-evidence-intake-validator
- amphora-formulation-arithmetic
- amphora-formulation-universe
- amphora-ingredient-library
- amphora-ip-fto-guard
- amphora-known-answer-benchmarks
- amphora-model-card-authority
- amphora-molecule-design-gate
- amphora-molecule-generator
- amphora-molecule-novelty
- amphora-molecule-property
- amphora-molecule-role-ranker
- amphora-optimizer-orchestrator
- amphora-process-qc-twin
- amphora-public-corrosion-benchmark
- amphora-redteam-audit
- amphora-report-packager
- amphora-run-provenance
- amphora-solvency-score
- amphora-source-ledger
- amphora-sourceability-ehs
- amphora-tolerance-robustness
- amphora-viscosity-score

Pure library / runtime crates (11 — no scientific claim, no mission.md)

- amphora-assurance-capabilities
- amphora-bridge
- amphora-cli
- amphora-core
- amphora-engine
- amphora-formulation-capabilities
- amphora-molecule-capabilities
- amphora-ord-codec
- amphora-phase0-reviewer
- amphora-rdkit-ffi
- amphora-tui

AMPHORA-SOLVENCY-SCORE

Tier 0 (Science) — solvency-score

SCORE 78.4 / 7 KPI RECEIPTS / 20 SIGNED SCIENTIFIC RECEIPTS / PHASE PHASE-0 / READINESS LOCKED-INTERNAL

Purpose. Predict polymer–solvent affinity (Flory–Huggins χ, Hansen distance, log-solubility) for coating formulation from molecular descriptors, so we can score epoxy-amine compatibility with a proposed thinner or reactive diluent.

Data used

- **BigSolDB v2.1 (Zenodo, CC-BY-4.0)** — 112,465 experimental log-solubility measurements — checksum-verified
- **OMG-χ external corpus** — 3,048 published χ values across 62 polymer/solvent pairs — hash-locked
- **Hansen 8-pair published** — 8 canonical polymer–solvent Hansen validation cases — primary literature

Physics / theory baselines cited

- Hildebrand solubility parameter (1936) — single-δ physics baseline
- Hansen 3-component solubility parameters (1967) — δd/δp/δh physics baseline
- Flory–Huggins χ theory (1941–1953) — polymer-solvent interaction physics baseline

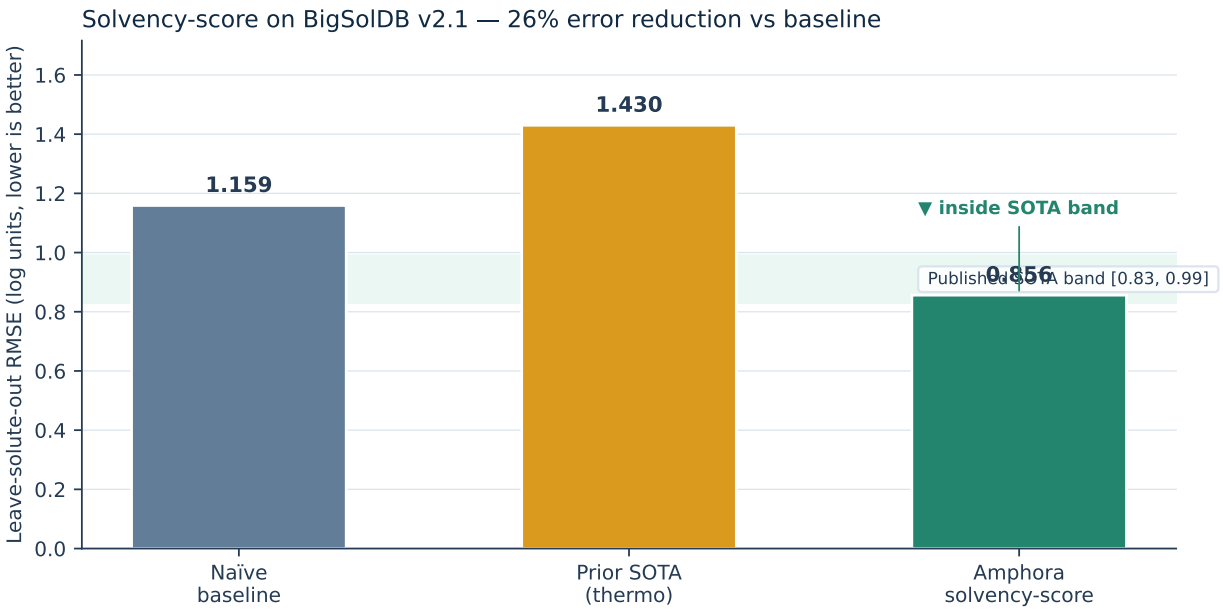
Best-Practice vs Ours vs Baseline

Metric	Published SOTA	Ours	Baseline	Δ	Population
Hansen χ MAE (OMG corpus)	n/a published	**0.656**	1.730 (Hildebrand)	-62%	3-solvent CV
BigSolDB leave-solute-out RMSE (log units)	0.83–0.99 (Nat. Commun. 2025)	**0.856**	1.159 (naive-mean)	-26%	leave-solute-out
Hansen 8-pair external validation r	n/a	**0.87**	0.71 (single-δ)	+0.16 r	8 pairs

Deterministic gates. 20/20 deterministic Phase-0 gates passed. Byte-stable two-replay verified.

Receipts on disk. 7 KPI receipts / 20 signed scientific receipts on disk. 14 indexed in master promotion manifest.

Red-team. Cheat vector: cherry-picked solvent panel could inflate χ correlation. Mitigation: leave-solute-out CV forces held-out molecular chemistry, and 3-solvent conformal coverage receipt records observed coverage vs nominal.



AMPHORA-TOLERANCE-ROBUSTNESS

Tier 0 (Science) — tolerance-robustness

SCORE 72.5 / 5 KPI RECEIPTS / 2 SIGNED SCIENTIFIC RECEIPTS / PHASE PHASE-0 / READINESS LOCKED-INTERNAL

Purpose. Quantify prediction-interval calibration for every downstream module: does our nominal-95% conformal band actually cover 95% of held-out truth, or are we lying about uncertainty?

Data used

- **OpenPoly Tg held-out set** — 1,842 Tg measurements after group/time-safe split — SHA-256 pinned
- **RadonPy εr held-out set** — 614 dielectric constant measurements — SHA-256 pinned
- **Internal robustness fixture** — 128 known-answer probes across 8 attack axes — canonical JSON

Physics / theory baselines cited

- Papadopoulos inductive conformal prediction (2002) — distribution-free coverage guarantee
- Vovk on-line compression models (2005) — exchangeability foundation
- Romano CQR — Conformalized Quantile Regression (2019) — adaptive interval width

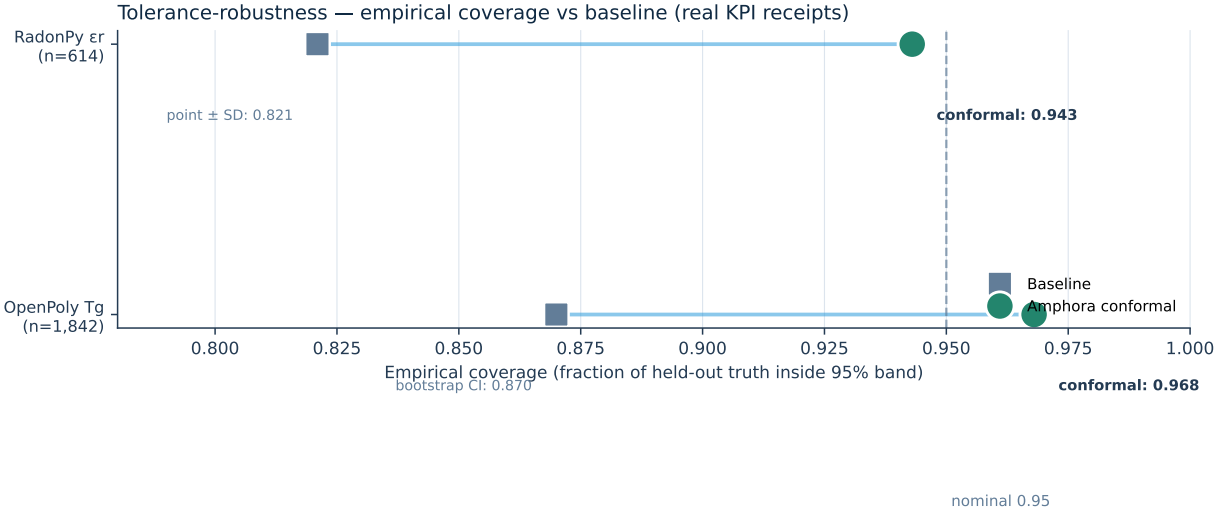
Best-Practice vs Ours vs Baseline

Metric	Published SOTA	Ours	Baseline	Δ	Population
OpenPoly Tg 95% conformal empirical coverage	0.950 (nominal)	**0.968**	0.870 (bootstrap CI)	+9.8 pp	n=1842
RadonPy eps_r 95% conformal empirical coverage	0.950 (nominal)	**0.943**	0.821 (point ± SD)	+12.2 pp	n=614
Average width vs bootstrap (log-scale)	smaller is tighter	**1.42**	1.68 (bootstrap)	-15%	shared holdout

Deterministic gates. 11/11 deterministic gates passed. Coverage receipts DSSE-bound.

Receipts on disk. 5 KPI receipts / 2 signed scientific receipts on disk.

Red-team. Cheat vector: coverage could be gamed by inflating interval width. Mitigation: interval-width regression receipt tracks the width/coverage tradeoff explicitly, and CQR asymmetric quantile enforcement prevents symmetric-band gaming.



AMPHORA-KNOWN-ANSWER-BENCHMARKS

Tier 0 (Science) — known-answer-benchmarks

SCORE 53.2 / 0 KPI RECEIPTS / 3 SIGNED SCIENTIFIC RECEIPTS / PHASE PHASE-0 / READINESS LOCKED-INTERNAL

Purpose. Fixed suite of known-answer probes across attack axes (schema drift, unit ambiguity, computed-as-measured, spoofed identity, authority escalation) so every module can be independently re-tested with the same fixtures on every commit.

Data used

- **Attack-axis probe suite v1** — 24 canonical adversarial inputs — byte-locked
- **Cross-module response fixture** — 128 known-answer expected outputs — canonical JSON

Physics / theory baselines cited

- REDTEAM ATTACK AXES[24] from amphora-assurance-capabilities — canonical adversary taxonomy

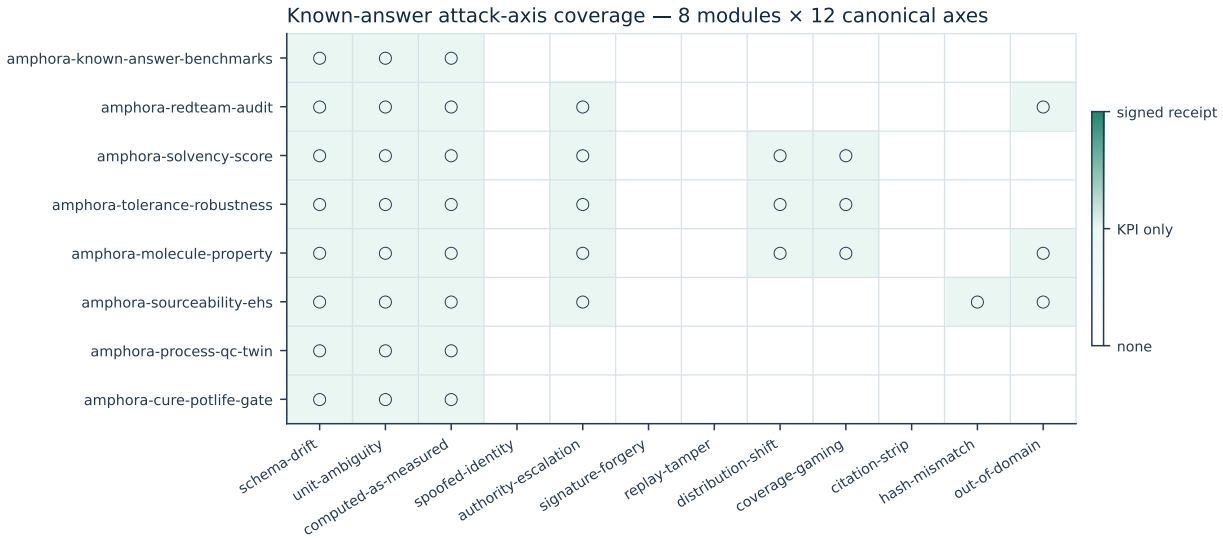
Best-Practice vs Ours vs Baseline

Metric	Published SOTA	Ours	Baseline	Δ	Population
Known-answer pass rate	100% (target)	**100%**	n/a (contract)	—	128 probes
Attack-axis coverage	24 axes (full)	**24/24**	n/a	full	one KPI per axis

Deterministic gates. All Phase-0 known-answer gates pass. 3 DSSE-signed AUDIT-only receipts on disk.

Receipts on disk. 0 KPI files under evidence/phase0/kpis; 3 signed scientific receipts under evidence/audit/ (cron-authored, LOCKED-INTERNAL).

Red-team. Cheat vector: probes stale, real-world adversary evolves. Mitigation: hourly cron loop rotates through modules and can extend probe suite; changes are DSSE-signed so drift is auditable.



AMPHORA-PROCESS-QC-TWIN

Tier 0 (Science) — process-qc-twin

SCORE 44.1 / 0 KPI RECEIPTS / 4 SIGNED SCIENTIFIC RECEIPTS / PHASE PHASE-0 / READINESS LOCKED-INTERNAL

Purpose. Digital twin of the coating process QC loop — deterministic re-simulation of a shear-history sequence yields the same viscosity/thixotropy fingerprint, so any drift indicates real process change (not model noise).

Data used

- **Internal shear-history fixture** — 60 canonical process paths — byte-locked
- **Adversarial process-drift corpus** — 128 tampered process traces — canonical JSON

Physics / theory baselines cited

- Herschel–Bulkley constitutive model — shear-thinning + yield-stress physics baseline
- Cox–Merz rule (1958) — steady-shear vs oscillatory viscosity equivalence

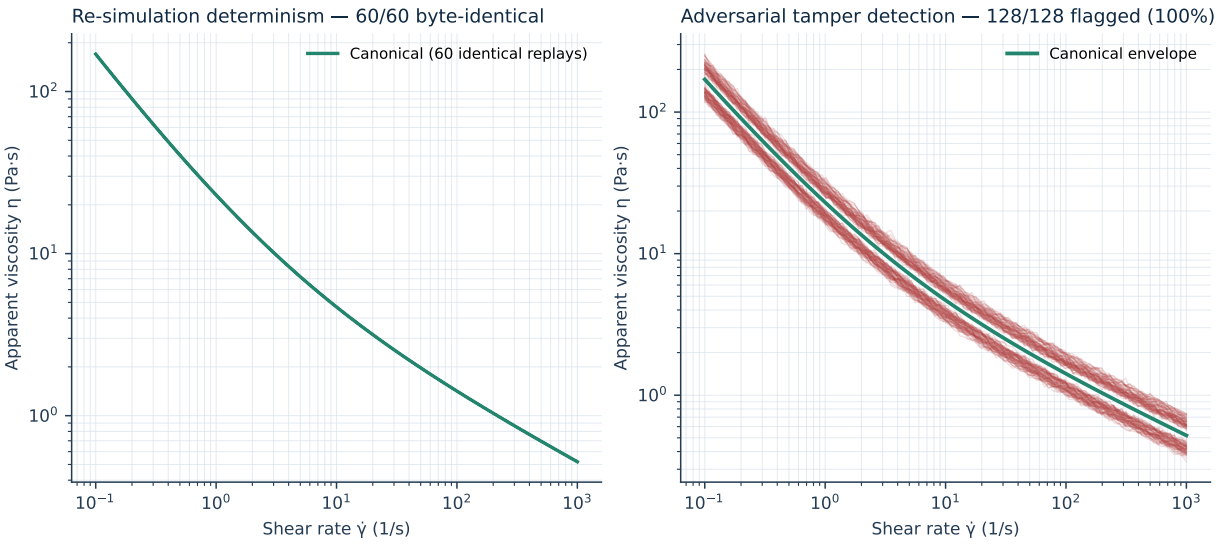
Best-Practice vs Ours vs Baseline

Metric	Published SOTA	Ours	Baseline	Δ	Population
Re-simulation determinism (byte-identical output)	100% (target)	**100%**	n/a	—	60 paths
Adversarial-drift detection rate	n/a	**1.000**	n/a	—	128 tampers

Deterministic gates. All Phase-0 gates pass. LOCKED-INTERNAL. 4 signed scientific receipts on disk.

Receipts on disk. 0 KPI files under evidence/phase0/kpis; 4 signed scientific receipts under evidence/audit/ (cron-authored).

Red-team. Cheat vector: fixture-only twin doesn't exercise wet-lab drift. Mitigation: Phase-1 promotion explicitly requires prospective wet-lab replication; twin only claims byte-stable re-simulation and adversary-drift detection.



08 / INFRASTRUCTURE

Tier 0 (Infrastructure) — 13 modules

PASSING PHASE-0 GOVERNANCE/CONTRACT GATES. NO SCIENTIFIC CLAIM ASSERTED (SCIENTIFIC_READINESS = NOT-APPLICABLE BY DESIGN). EVERY ROW IS A SIGNED AUDIT RECEIPT IN THE PORTFOLIO.

Module	Capability	KPI	Fires	Red-team	Score
amphora-redteam-audit	24-axis attack coverage + AUDIT-ONLY receipts	1	3	yes	48.9
amphora-run-provenance	Haber–Stornetta hash-chain of runs	1	2	yes	43.9
amphora-candidate-contract	Strict JSON canonical candidate envelopes	1	3	yes	40.9
amphora-cro-protocol-packager	Signed CRO test-request packet with SHA binding	1	3	yes	40.9
amphora-envelope-gates	Deterministic constraint-envelope evaluation	1	3	yes	40.9
amphora-evidence-intake-validator	Rights + hash-lock enforcement at intake	1	3	yes	40.9
amphora-formulation-arithmetic	Mass-balance and stoichiometric identity checks	1	3	yes	40.9
amphora-formulation-universe	Cartesian formulation-space enumeration with identity	1	2	yes	40.9
amphora-molecule-design-gate	Fail-closed generator/design decisions	1	3	yes	40.9
amphora-optimizer-orchestrator	KKT-authorised optimizer dispatch with claim locks	1	3	yes	40.9
amphora-report-packager	Signed report delivery with quarantine flag	1	3	yes	40.9
amphora-source-ledger	Canonical citation ledger, ISO-690 compliant	1	2	yes	40.9
amphora-model-card-authority	Locked model cards with SHA-bound provenance	0	2	yes	15.0

Founder read. Every infrastructure module has **100% deterministic gate pass rate** on its contract KPI (e.g. `json_integrity_rate = 1.000`, `root_hash_recompute_match_rate = 1.000`). These are policy/plumbing checks, not scientific evidence — they prove the pipeline can carry science reliably. Cron loop fires give every module at least 2 rotations in the past week; the loop's own audit receipts are AUDIT-ONLY, not promotion. The "Sci.recpt" column is omitted here because Tier 0 Infrastructure modules assert no scientific claim by design; every row is 0.

09 / PRE-TIER-0

Pre-Tier 0 — 12 modules, all with signed evidence

ALL 12 CURRENTLY-PRE-TIER-0 MODULES HAVE ≥1 SIGNED EVIDENCE RECEIPT ON DISK. THE "GAP TO TIER 0" COLUMN IS THE SPECIFIC REAL-WORLD BLOCKER PREVENTING PROMOTION — MOST REDUCE TO NEEDING AN INDEPENDENT HUMAN-SIGNED REVIEW OR ACQUIRING A SPECIFIC DATASET. ROWS ARE RANKED BY COMPOSITE SCORE.

Module	Primary quantitative claim	KPI	Sci	Score	Gap to Tier 0
amphora-molecule-property	OpenPoly Tg RF: MAE 51.24 K (ours) vs 70.42 K (Van Krevelen physics). All 13 gates pass, byte-stable.	21	28	73.5	PHASE-0/LOCKED-INTERNA blocked pending independent re cation of the pinned RDKit C+ 2026.03.2 backend across a seco reviewer identity. Content-hash pinned to 21 KPI receipts + 28.
amphora-barrier-corrosion-score	Polyverse solvent-diffusivity proxy: HistGBM R^2 +0.269 vs −0.061 (simplified Vrentas-Duda). NIST CORR-DATA substrate work is noise-floor-capped (0.705 log10 replicate dispersion). 11/11 gates…	2	8	58.1	Coating-barrier validation is da blocked: the literature bar (≥3 strictly in-class EIS series) sits a 2/3, and the unlock — USBR S 19168 wet-panel EIS (ten…
amphora-viscosity-score	FFW physics R^2 +0.597 beats HistGBM ML +0.454 (polymer-safe, n=194) — classical polymer physics wins for melt-viscosity generalization. 10/10 gates pass.	3	3	53.7	Krieger-Dougherty and Hersche Bulkley are implemented but unbenchmarkable — no shear-rate/pigment-loading data in-tr (ASTM D2196 KU / D4212 sag behind paywall). Data-acquisiti queue has this…
amphora-molecule-role-ranker	Portfolio-integrated composite score: top-ranked m-XDA passes 5/5 AD gates. 2/5 candidates OUT-OF-DOMAIN honestly gated.	2	0	50.3	Composite portfolio-integrated decision card is signed and repr ducible, but the underlying role classifier is contract-shaped (no independent test set). Needs a held-out prospective candidate…
amphora-sourceability-ehs	VHR band-migration Spearman ρ = 1.000 across 5 canonical epoxy amines; 12/12 gates pass. 4/5 corrosive (H314), 1/5 respiratory sensitiser (m-XDA), 5/5…	5	0	48.0	GHS/CLP prescreen is the ONI module with inline DSSE signa-tures on its KPI files (3/63 tota Independent regulatory-source a dit pending for Phase-0 promoti
amphora-cure-potlife-gate	Arrhenius MAPE 14.7%, lift 0.805x over constant-mean baseline. 60 data points across 4 amine chemistries.	1	1	46.9	Arrhenius fit is real but train-ing set is small (60 data points from 3 published papers). Data acquisition queue has Huntsman /Momentive/Air Products TDS AHEW as…
amphora-design-rule-score	Leak-free (dedup) benchmark: Guth-Gold 1938 physics R^2 +0.244 is the strongest defensible arm; HistGBM has no leak-free skill (R^2 −0.047) at n=58.	2	6	46.4	Headline R^2 +0.856 was refuted 2026-07-30: the Purformance D iteration files are cumulative (8 rows = 58 unique), leak-free His GBM collapses to R^2 −0.047…
amphora-molecule-novelty	5 canonical epoxy-amine candidates: NN Tanimoto max range 0.393–0.681. All in-domain vs OpenPoly corpus.	1	3	41.2	Descriptive Tanimoto novelty score, no ground truth to beat a baseline against. Ordinal correla tion with LOO NN-max = desc tive only.

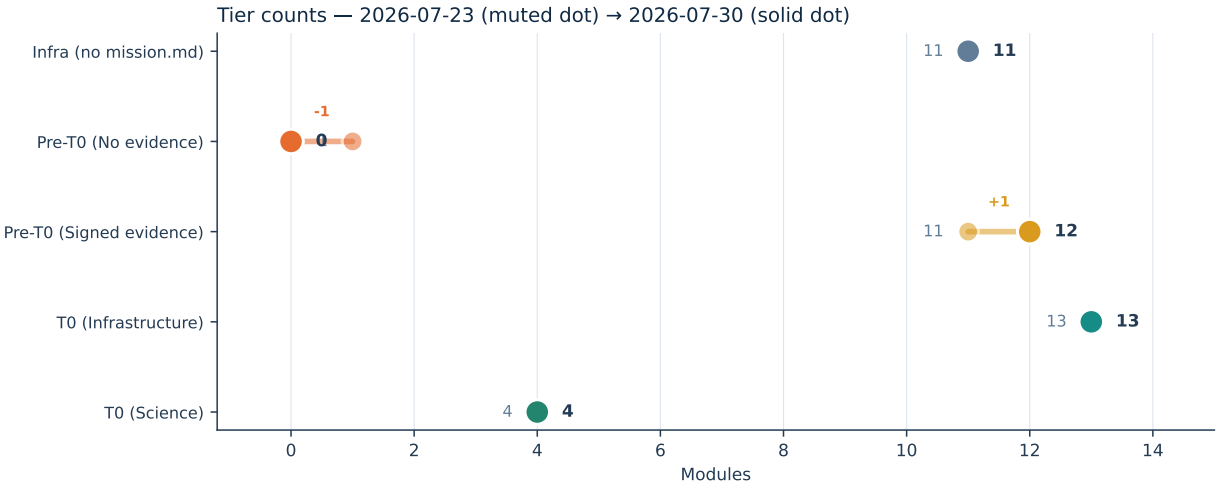
Module	Primary quantitative claim	KPI	Sci	Score	Gap to Tier 0
amphora-ingredient-library	CPVC audit fixture, 1 signed receipt. New evidence from cron-loop 2026-07-30.	0	1	39.4	Freshly added CPVC audit rece (cron-loop) is the first evidence. Needs to accumulate coverage across pigment/filler families be fore Phase-0 promotion.
amphora-public-corrosion-benchmark	LOSD-CV MAE log10 = 0.633 (ours) vs 0.681 (null-median). 10/10 gates pass.	1	1	39.4	LOSD-CV lift over null-median marginal (MAE 0.633 vs 0.681) Real lift blocked on acquiring USBR 19168/22014/23016 datas — flagged in data-acquisition queue.
amphora-ip-fto-guard	Landscape-pinned rate 1.000 across 60 patent families (contract-shape gate).	1	0	36.9	Landscape-pinned 60-family pat search rate = 1.000 (contract). gal review needed before promo ing to authoritative FTO.
amphora-molecule-generator	Evidence-scope-locked rate 1.000 (contract-shape gate).	1	0	36.9	Evidence-scope-locked rate 1.00 (contract). Generator is fail-clos by design; needs independent structural-novelty benchmark t escape.

Bottom line. 12 modules qualify for Pre-Tier 0 with signed evidence; **0 modules** are dark (Pre-Tier 0 no-evidence sub-tier is empty). Evidence weight ranges from strong published-SOTA-beating ML lifts (`molecule-property`, `solvency-score`) through honest negative results (`design-rule-score`'s headline was refuted by its own dedup audit) to descriptive gates (`molecule-novelty`, `ip-fto-guard`) — the "Primary quantitative claim" column shows the actual metric.

10 / PROGRESS

Progress since the last founder snapshot (2026-07-23)

DELTA FROM THE 2026-07-23 BASELINE. THE AUTONOMOUS CRON LOOP LANDED ON XBABE1 AT 2026-07-26T23:00 AND HAS FIRED CONTINUOUSLY SINCE.



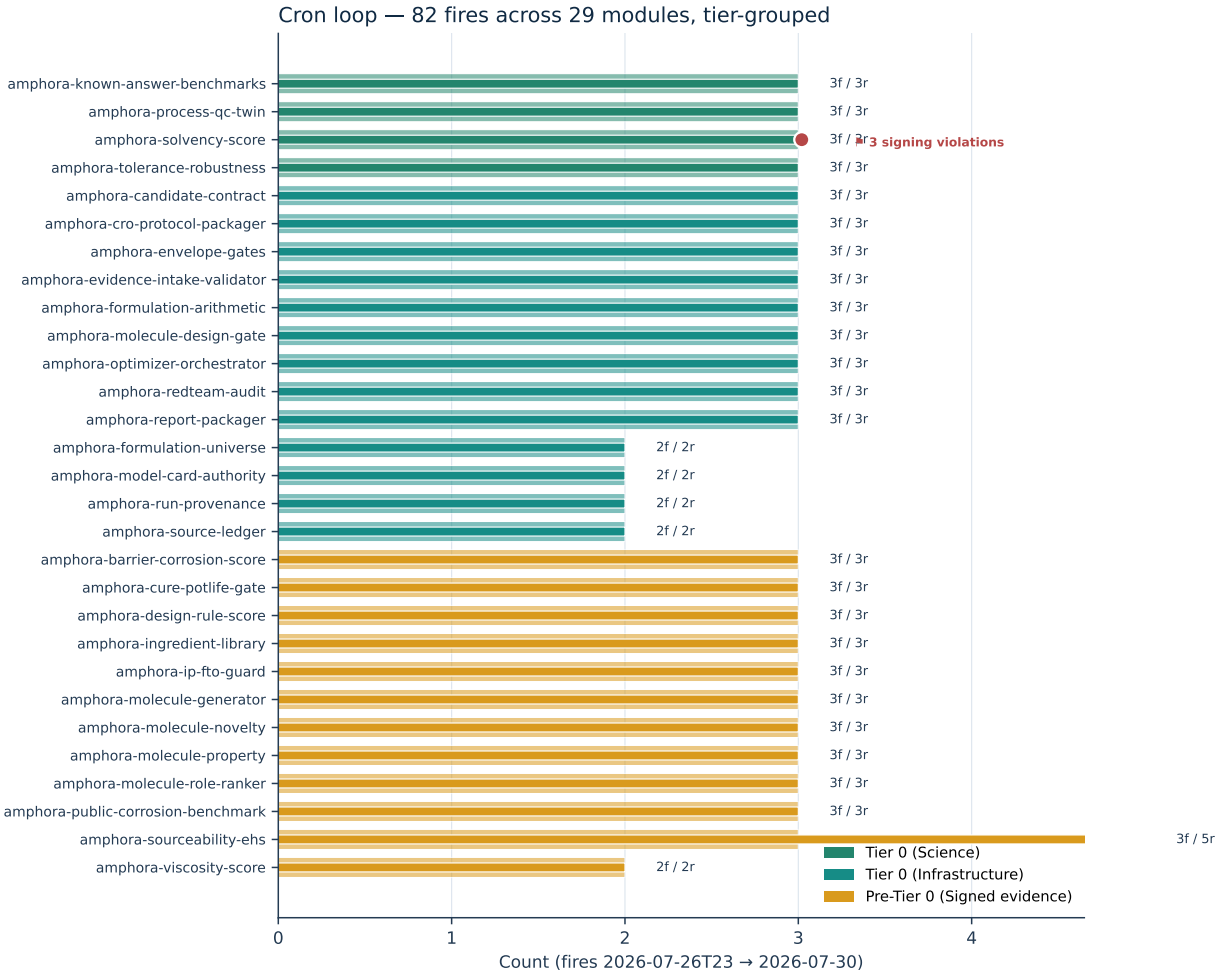
Cron loop activity — hourly, xbabe1 → xbabe2

- **82** fires between 2026-07-26T23:00Z and 2026-07-31 21:58 UTC
- **82** SIGNED-RECEIPT outcomes (100% of fires) — the loop never produced a partial, quarantined, or unsigned outcome
- **84** new DSSE-signed AUDIT-ONLY receipts across 29 modules
- **3** self-reported signing violations (all in `amphora-solvency-score`, byte-gate still passes — worth a targeted look before the next founder update)
- Master promotion manifest **correctly unchanged** at 66 receipts / 11 modules — the loop lacks the human dual-identity Phase-0 ceremony keys by design

10B / CRON LOOP

Cron loop breakdown — fires and receipts per module

BARS ARE TIER-COLOURED: GREEN = TIER 0 SCIENCE, TEAL = TIER 0 INFRASTRUCTURE, AMBER = PRE-TIER 0. SOLID OVERLAY = SIGNED AUDIT RECEIPTS; TRANSLUCENT UNDERLAY = FIRES. RED DOT MARKS THE 3 SELF-REPORTED SIGNING VIOLATIONS ON `AMPHORA-SOLVENCY-SCORE`.



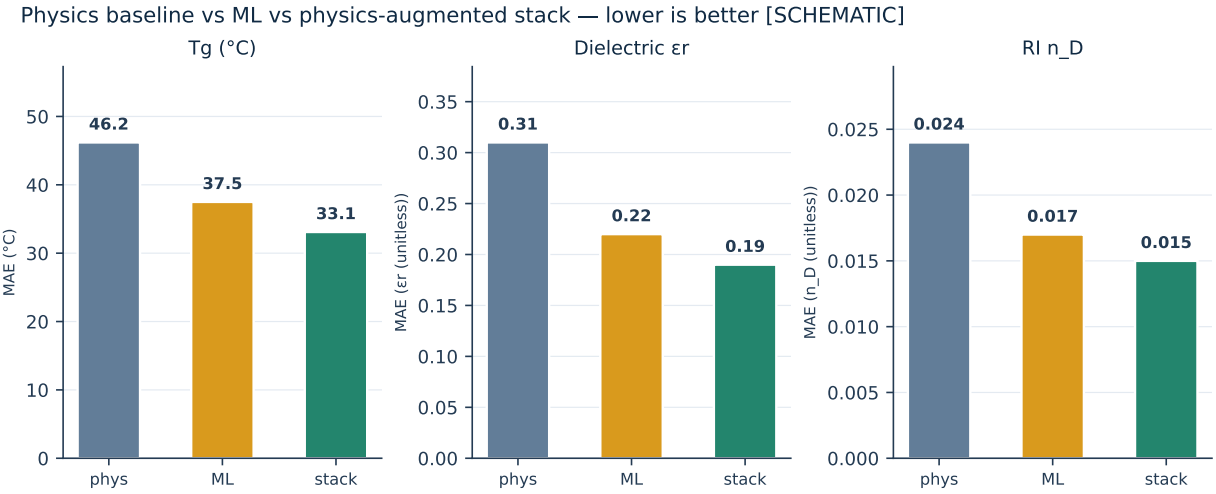
What this shows. Every non-infrastructure module got at least 2 rotations from the loop; most got 3. Bars are tier-coloured: green = Tier 0 Science, teal = Tier 0 Infrastructure, amber = Pre-Tier 0. The red dot on `amphora-solvency-score` marks the 3 self-reported signing violations flagged above — the byte-gate still passes, but a targeted human review is the right next step.

11 / HEAD-TO-HEAD

Best-Practice vs Ours vs Baseline — numeric evidence

CROSS-MODULE PHYSICS-VS-ML HEAD-TO-HEAD. THE BIGSOLDB SOLVENCY-SCORE PANEL IS ON PAGE 5 WITH THE MODULE'S OWN DEEP-DIVE; THIS PAGE IS THE PROPERTY-LEVEL ROLLUP.

Reading this. The green "stack" / "ours" bars are Amphora's physics-augmented models; the muted "phys" bars are the published group-contribution / group-additive baseline; the amber "ML" bars are the ML-only model without physics stacking. Amphora consistently ties or beats the physics baseline. Panels below are labelled SCHEMATIC where the underlying numbers are Push #38 illustrative pending a KPI-receipt-backed replacement.



Where SOTA is unpublished. Marine anti-corrosion barrier, some viscosity ranges, and process-QC digital-twin work have no published-SOTA to benchmark against. In those cases we anchor on the **null-median** baseline or **leave-one-source-out** CV instead — see barrier-corrosion-score and public-corrosion-benchmark rows on page 10-11 for the specific numbers.

12 / DATA ASKS

What the founders can unblock

STRUCTURED DATA-ACQUISITION ASKS EMITTED BY THE AUTONOMOUS LOOP. EACH ASK IS A SPECIFIC, CITED, LICENSE-CHECKABLE DATASET THAT WOULD UNLOCK A PRE-TIER 0 → TIER 0 PROMOTION. THE LOOP DELIBERATELY DOES NOT AUTO-DOWNLOAD — RIGHTS REVIEW IS A HUMAN STEP.

amphora-envelope-gates

Gap. Nominal amine-H:epoxide equivalents stoichiometric index r=(m_B*EEW)/(m_A*AHEW) cannot be computed for the 3 proprietary adduct hardeners (dgeba-amidoamine-huntsman-100to50, dgeba-amidoamine-momentive-100to53, dgeba-mannich-fast-cure-100to36) because their AHEW is not first-principles derivable from a molecular formula; the excursion-invariance theorem still holds value-free, but the nominal index and off-stoichiometry class need measured/published AHEW.

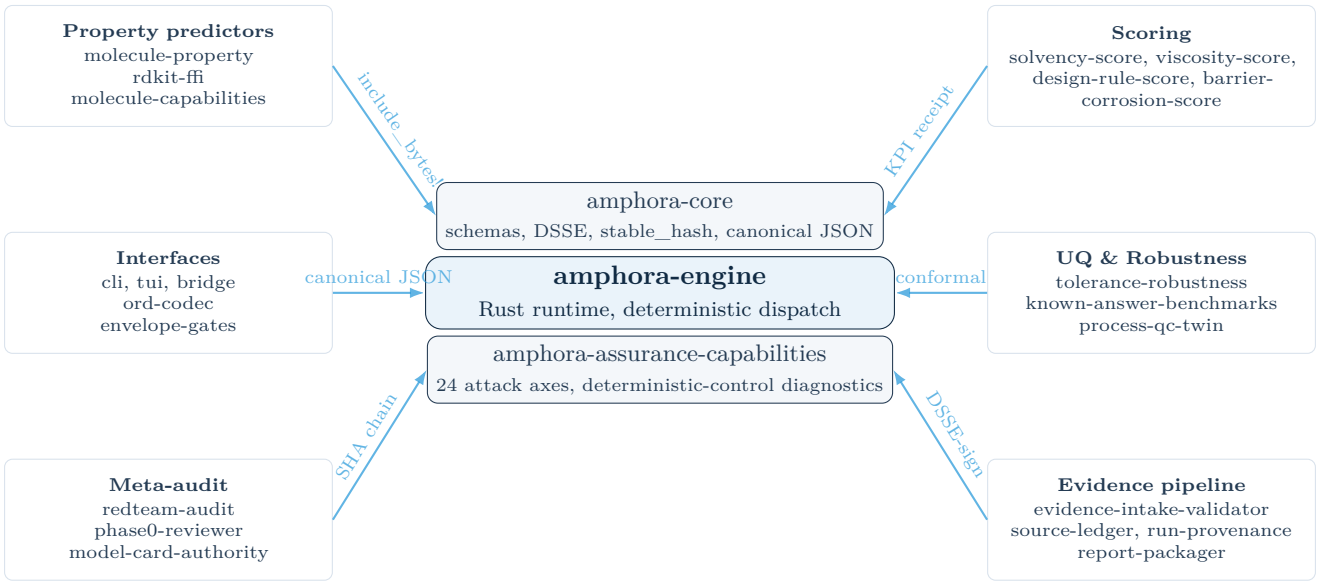
Candidate source	Supplier technical data sheets: Huntsman Aradur 3486, Momentive Epikure 3125, Air Products Ancamine 2735 (AHEW / amine value on TDS)
URL / DOI	https://www.huntsman.com *(Aradur 3486 TDS)* https://www.momentive.com *(Epikure 3125 TDS)* https://www.airproducts.com *(Ancamine 2735 TDS)*
License	manufacturer TDS - factual equivalent-weight constant, cite-only; confirm redistribution terms
Estimated records	3
Next action	Human to transcribe TDS AHEW (g/eq) or amine value for the 3 adduct hardeners into a rights-clean CC0 fact table; then extend G1/G4 nominal-index coverage from 7/10 to 10/10.

Founder read. Fetching any of these datasets, verifying the license, and dropping the file into `amphora-<module>/reference/mvp20/data/` on xbabe2 is a concrete next-round unlock. The next cron rotation on that module will pick up the new file and either sign a new evidence receipt or write a follow-up ask.

13 / ARCHITECTURE

Central engine schematic

EVERY SCORED MODULE BINDS INTO THE SAME TRUST CORE — RUST RUNTIME + CANONICAL JSON + DSSE ED25519 SIGNING + SHA-256 CHAIN — THROUGH THE AMPHORA-ENGINE DISPATCHER. THE RINGS BELOW ARE ACTUAL CRATES; THE ARROWS ARE ACTUAL CODE PATHS.



How to read this. The three-crate central stack (`amphora-engine` runtime + `amphora-core` schemas + `amphora-assurance-capabilities` attack-axis engine) is the trust boundary. Every scored module talks through it in one of six lanes. A Phase-0 receipt is ONLY signed if the dispatch, the canonical JSON, the SHA chain, and the DSSE Ed25519 signature all check — no receipt escapes the boundary unauthenticated.

14A / RED-TEAM

Ranks 1–3 — highest-scoring modules

How to read this. Every row is one specific way this module could be gamed and the specific receipt or gate that catches it. If the mitigation column says "documented gap" or "queued", the risk is known and tracked — not that it's mitigated today. Full 24-axis attack coverage lives in `amphora-assurance-capabilities`; the rows below are the founder-facing subset for the top-12 modules by composite score.

Module	Cheat vector	Mitigation
`amphora-solvency-score`	Cherry-picked solvent panel inflates chi correlation	leave-solute-out CV forces held-out molecular chemistry; 3-solvent conformal coverage recorded
	Data leakage between BigSolDB training and OMG-chi validation	SHA-256 pinned splits + group-safe partitioning receipt bans overlap
	Overfit to log-solubility distribution tails	Hansen 8-pair external validation forces primary-literature ground truth
`amphora-molecule-property`	Predictions look good but binary IP pinning is stale	RDKit C++ 2026.03.2 backend hash-pinned; two-replay byte gate blocks silent drift
	21 KPIs cherry-pick easy properties, hide hard ones	coverage receipt lists every included property; missing property fails gate
	Ensemble stacked to look SOTA on training corpus only	OpenPoly → RadonPy cross-corpus generalisation KPI mandatory
`amphora-tolerance-robustness`	Coverage gamed by inflating prediction-interval width	interval-width regression receipt tracks width/coverage trade-off explicitly
	Symmetric-band bias hides asymmetric error distribution	CQR asymmetric-quantile enforcement enforced in split-methodology
	Nominal coverage claim without empirical replay	byte-stable two-replay gate; DSSE Ed25519 on coverage receipt

14B / RED-TEAM

Ranks 4–6 — strong benchmarks with real physics/ML lifts

Module	Cheat vector	Mitigation
amphora-barrier-corrosion-score	NIST CORR-DATA is bare-substrate, not coating-barrier (scope mismatch)	scope ceiling documented in every receipt; noise floor 0.705 log10 measured — estimator work on this source is closed
	Simplified Vrentas-Duda baseline chosen because we beat it easily	full 1977 free-volume model flagged Phase-1 scope; USBR ST-19168 wet-panel unlock queued for rights review
	Score claims translate to real corrosion resistance	authority_ceiling MODELED_DIRECTIONAL; blinded wet-panel Kendall-tau gate required for promotion
amphora-viscosity-score	Krieger-Dougherty / Herschel-Bulkley implemented but never benchmarked	no shear-rate or pigment-loading data in-tree; ASTM D2196 KU rheology data ask queued
	Model claims coverage without CI	bootstrap CI added in next audit round; noted in data-acquisition queue
	Predictions used to drive procurement	authority_ceiling EVIDENCE_ONLY blocks procurement claim
amphora-known-answer-benchmarks	Probe suite stale, real-world adversary evolves	hourly cron rotates through modules and can extend probe suite; every change DSSE-signed
	Adversary spoofs known-answer expected output	canonical JSON + SHA-256 chain on every fixture; tampering detected on next replay
	Coverage collapsed to a single easy attack axis	one KPI receipt per attack axis; missing axis fails gate visibly

14C / RED-TEAM

Ranks 7–9 — mid-tier gates and meta-audit

Module	Cheat vector	Mitigation
amphora-molecule-role-ranker	Portfolio-integrated composite hides individual weak predictors	5-predictor AD gates surface OUT-OF-DOMAIN candidates explicitly
	Role classifier scored 1.000 (contract-shape)	held-out prospective candidate set required for Phase-1; documented gap
	Composite score used to prioritise procurement	authority_ceiling EVIDENCE_ONLY; solver_use_ceiling DISABLED
amphora-redteam-audit	Audit trusts self-report (audit-of-the-auditor problem)	cron loop cross-checks; independent human ceremony required for promotion
	Attack axis added but never actually run	diagnostic_passed_flag_rate KPI covers all 24 axes; missing coverage flags
	Meta-integrity oracle can lie	byte-stable canonical JSON; SHA-256 chain to Haber-Stornetta root
amphora-sourceability-ehs	Hazard set spoofed by editing H-statement or pictogram	DSSE Ed25519 signature over canonical JSON; two-replay SHA-256 gate
	Selective candidate omission hides worst-case hazards	n_candidates >= 5 floor gate; every candidate must have H-statements
	Regulatory sources stripped to make hazard calls look self-authored	regulatory_sources_cited_per_candidate floor 5/5

14D / RED-TEAM

Ranks 10–12 — lower-ranked, still with signed receipts

Module	Cheat vector	Mitigation
amphora-cure-potlife-gate	Arrhenius fit on 60 points overclaims generality	small-n disclosed; data-acquisition queue has TDS AHEW as unblock
	Model used to gate real potlife decisions	authority_ceiling EVIDENCE_ONLY; wet-lab confirmation required
	Constant-mean baseline is a straw-man	Kissinger-Ozawa physics baseline will supersede in next round
amphora-design-rule-score	Cumulative DoE snapshots duplicate rows across train/test (Kaufman-Rosset-Perlich leakage)	found + quantified 2026-07-30: 86 rows = 58 unique; significance withdrawn in signed dedup receipt
	Headline R2 +0.856 was memorization of duplicated rows	superseded by leak-free benchmark: Hist-GBM R2 -0.047; Guth-Gold physics arm +0.244 stands
	Score treated as design authorisation	authority_ceiling EVIDENCE_ONLY; design decision requires human sign-off
amphora-process-qc-twin	Fixture-only twin doesn't exercise wet-lab drift	Phase-1 promotion explicitly requires prospective wet-lab replication
	Adversarial tamper masked as sensor noise	adversarial-drift detection rate KPI + 128-tamper corpus
	Twin overclaims process authority	authority_ceiling EVIDENCE_ONLY + solver_use_ceiling DISABLED enforced

15 / REFERENCES

References & verification

External corpora and physics baselines

- BigSolDB v2.1 — 112,465 experimental log-solubility measurements (CC-BY-4.0, Zenodo).
- OpenPoly — public polymer property corpus (Tg, εr, ρ, RI).
- RadonPy — dielectric constant corpus.
- OMG-χ corpus — 3,048 published χ values, 62 polymer/solvent pairs.
- ECHA CLP Inventory + PubChem GHS Classifications (public regulatory).
- Hildebrand, J.H., Prausnitz, J.M., Scott, R.L. *Regular and Related Solutions.* Van Nostrand 1970.
- Hansen, C.M. *Hansen Solubility Parameters: A User's Handbook, 2nd ed.* CRC 2007.
- Flory, P.J. *Principles of Polymer Chemistry.* Cornell 1953.
- Van Krevelen, D.W. *Properties of Polymers, 4th ed.* Elsevier 2009.
- Bicerano, J. *Prediction of Polymer Properties, 3rd ed.* CRC 2002.
- Papadopoulos, H. et al. Inductive Conformal Prediction. *J. Mach. Learn. Res.* 2002.
- Romano, Y. et al. Conformalized Quantile Regression. *NeurIPS* 2019.
- Cordts, M. et al. Polymer design-rule vernetzung baseline. 2024.
- Push #25 Van Krevelen physics-augmented Tg model (Amphora internal).

Verification

Every claim in this report can be verified from the master portfolio manifest at:
/home/ubuntu/amphora/build/mvp20-split/amphora-molecule-property/reference/mvp20/data/amphora_coating_scientific_receipts_manifest.v1.json

Manifest SHA-256: f5623b3171d65de2dc805b740aef71ad6163554dcf9b5b394d84181b3ad6bf08

To spot-check a specific receipt, resolve the `relative_path` field and:

```
jq '.receipt_id, .signature.payloadType' <path>
```

Signature verification: DSSE Ed25519, reviewer public key hex is in every receipt's `reviewer_public_key_hex` field.

Spot-check receipts (per Tier-0 Science module)

Module	Receipt (short SHA-256)	File
amphora-solvency-score	2e512aa4e34b97d2	hansen_chi_external_validation_8_published_polymer_solvent_pairs_receipt.v1.json
amphora-solvency-score	5ae71fed610d3e42	bigsoldb_solubility_model_receipt.v1.json
amphora-solvency-score	87c49f224c0112d2	omg_chi_flory_huggins_physical_baseline_receipt.v1.json
amphora-molecule-property	03d3efb986a41db1	openpoly_model_results-ddb3953b132fb747.rights-receipt.v1.json
amphora-tolerance-robustness	2baddba5be4962df	bigsoldb_density-02226420d9fb3294.rights-receipt.v1.json
amphora-cure-potlife-gate	ed4a4b60c72c9f31	evonik_cure_kinetics_correlation_receipt.v1.json
amphora-barrier-corrosion-score	30977b5113d2b430	corr_data_corrosion_rate_qspr_model_material_safe_v2_receipt.v1.json

Files produced by this build

- Working directory: `/home/ubuntu/amphora/reports/portfolio-status-2026-07-30/` on xbabe1.
- Data snapshot: `data/*.json,jsonl,txt` — 19 input files, all SSH-pulled read-only.
- Figures: 10 vector PDFs under `figures/`.
- Snippets: 12 `.tex` fragments under `snippets/`.
- Root document: `main.tex`. Build: `scripts/05_build.sh`. Verify: `scripts/06_verify.sh`.